Track 1: Genomic Diagnosis of Mosaic Variegated Aneuploidy

Independent candidate-gene interrogation, BUB1B interpretation, and explicit uncertainty framework

Final molecular hypothesis
BUB1B-associated MVA1 is strongly supported, but not molecularly closed: p.Leu737Ter is an established pathogenic loss-of-function allele; p.Asn1002Lys remains a VUS/candidate second allele; phase is unresolved.

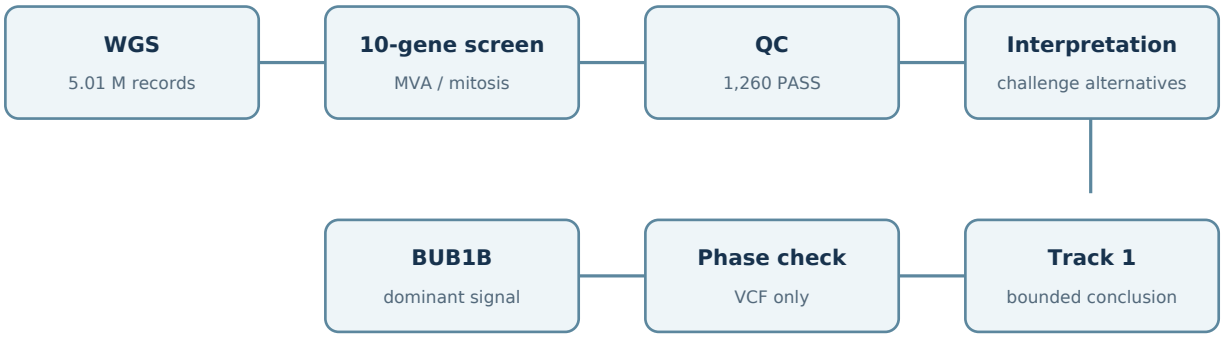
5,012,204	1,267	1,260	2
VCF records	candidate-locus	PASS	priority BUB1B calls

Boundary: research challenge interpretation, not a clinical diagnostic report. Raw genomic data are not reproduced.

Assembly: GRCh38. BUB1B transcript: NM_001211.6.

1. What was actually done

The full local WGS VCF was scanned once and reduced to a pre-specified chromosome-instability/MVA gene space before interpretation. The panel included BUB1B, CEP57, TRIP13, CENATAC, SLF2, SMC5, MAD1L1, BUB1, MAD2L1BP, and CEP192. Technical filtering produced 1,260 PASS variants; presence inside a locus was never treated as evidence of pathogenicity.



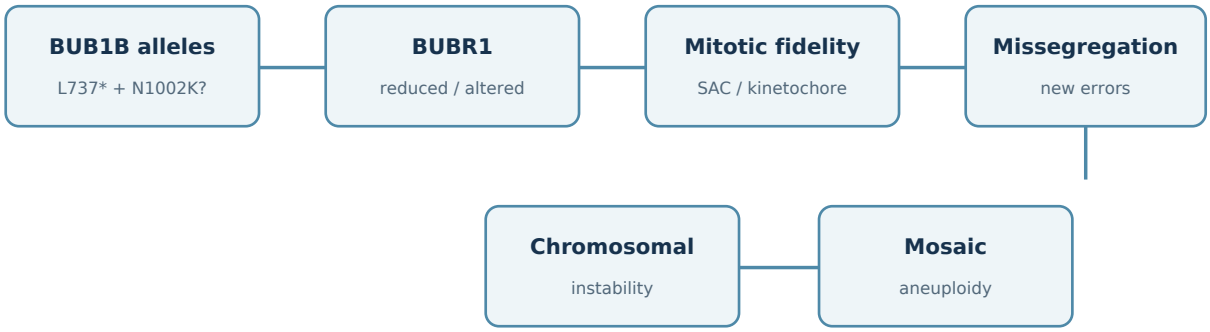
2. Established evidence vs working hypothesis

Evidence level	Finding	Interpretation
ESTABLISHED	BUB1B chr15:40209701 T>G; c.2210T>G; p.Leu737Ter; 0/1; DP46; GQ99	High-quality nonsense call. ClinVar: Pathogenic for MVA1.
SUPPORTED / UNRESOLVED	BUB1B chr15:40220612 T>G; c.3006T>G; p.Asn1002Lys; 0/1; DP28; GQ99	Technically strong missense call, but remains a VUS/candidate second allele.
UNRESOLVED	Relationship between the two alleles	No BAM/CRAM/FASTQ and no sufficient VCF phase information; cis vs trans cannot be established.
HYPOTHESIS	Biallelic BUB1B dysfunction as the patient's molecular mechanism	Strongly coherent with MVA, but dependent on the unresolved second allele and phase.

ClinVar nuance. p.Leu737Ter is currently classified Pathogenic for MVA1, but the record is based on criteria provided by a single submitter (Labcorp/Invitae), last evaluated October 9, 2024. It is strong external clinical evidence, but not an expert-panel assertion.

3. Why BUB1B remains the leading explanation

The strength is convergence rather than any single score: an established MVA gene, an exact pathogenic loss-of-function allele, a second rare high-quality missense candidate in the same gene, and a phenotype centered on mosaic aneuploidy. The alternative-gene screen was used as a falsification step rather than as a list of genes to force into the diagnosis.



The question mark is intentional: the mechanistic chain is compelling only if the second allele contributes to BUB1B dysfunction and the alleles form a disease-relevant configuration.

4. Alternative-gene challenge

The screen included established and emerging chromosome-instability genes rather than restricting the search to BUB1B. No alternative configuration identified in the current targeted analysis has comparable combined evidence. This is narrower than claiming that an exhaustive clinical genome analysis excluded every other diagnosis.

5. Phase analysis

A dedicated readiness check found 2 VCF files but no BAM, CRAM, alignment indexes, FASTQ, or local reference FASTA. Read-backed phasing therefore cannot be performed from the available dataset. Two heterozygous calls in one gene are not sufficient to infer trans configuration. Parental segregation or validated long-range/read-level data would be required.

6. Final interpretation

Leading molecular explanation: BUB1B-associated MVA1.

The patient carries a high-quality heterozygous pathogenic BUB1B loss-of-function variant, p.Leu737Ter, together with a second high-quality heterozygous missense variant, p.Asn1002Lys, that remains a VUS/candidate second allele. Phase is unresolved. The appropriate Track 1 conclusion is therefore candidate compound heterozygosity, not confirmed biallelic pathogenic BUB1B disease.

Highest-value evidence still missing

Parental segregation: establish cis/trans. Functional N1002K assay: test BUBR1 abundance/localization and checkpoint function. Cellular phenotype: chromosome alignment, lagging chromosomes, micronuclei, and missegregation. WT-BUB1B rescue: causal benchmark if patient cells are available.

7. How this differs from the public K-Dense Track 1

The public K-Dense submission independently converges on the same two BUB1B variants. That convergence is useful validation, but our report should not imitate its framing. K-Dense describes a phenotype-guided ranking workflow centered initially on BUB1B/CEP57/TRIP13, VEP annotation, a genome-wide ClinVar pathogenic/likely-pathogenic countercheck, and a ranked CSV with manually assigned confidence values. It also reports unresolved phase, N1002K as a VUS, and no successful independent Exomiser ranking.

Dimension	Our Track 1	K-Dense public Track 1
Primary framing	MVA/mitotic-fidelity hypothesis challenge across a 10-gene panel	Phenotype-guided variant prioritization and ranked candidate submission
Initial search space	10 pre-specified MVA / chromosome-segregation genes	Targeted established MVA review emphasized BUB1B, CEP57, TRIP13
Core differentiator	Alternative-gene falsification + explicit established-vs-hypothesis evidence model	Reproducible VEP/ranking pipeline + genome-wide ClinVar P/LP countercheck
BUB1B result	L737* pathogenic + N1002K VUS; candidate pair	Same leading pair
Phase	Explicit local readiness test: VCF present, no BAM/CRAM/FASTQ; unresolved	VCF lacks sufficient phase evidence; parental/long-range testing required
Emphasis	Mechanistic diagnosis boundary and genotype-to-Track-2 bridge	Auditable ranking, deterministic reproducibility, phenotype-guided curation

Strategic implication. The shared BUB1B finding is expected if it is the true biological signal. Our distinctive contribution is the falsification-oriented 10-gene screen, strict separation of established evidence from inference, explicit failed phase-resolution attempt, and genotype-to-mechanism bridge that directly motivates Track 2 experiments.

8. Track 1 -> Track 2 handoff

Track 2 should not assume that the genotype has been completely solved. It should ask whether residual BUBR1 function exists and whether a pharmacologic intervention can preserve or compensate for that residual function without worsening chromosome segregation.

Established	Unknown	Track 2 consequence
L737* pathogenic LoF	N1002K functional effect	Do not assume complete BUBR1 absence.
BUB1B-MVA mechanism is biologically coherent	cis vs trans	Drug hypothesis must remain valid under genotype uncertainty.
Missegregation is mechanistically relevant	Residual BUBR1 abundance/localization	Prioritize measurable cellular rescue endpoints, not symptom-only rationale.

9. References and provenance

NCBI ClinVar. NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter), RCV000641226.9. Pathogenic; criteria provided, single submitter; last evaluated Oct 9, 2024.

Hanks et al. Nature Genetics 2004. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. PMID 15475955.

K-Dense public Track 1 repository (2026). Used only as an external workflow comparison and independent convergence check; its ranking was not used as a substitute for our local candidate-locus analysis.

Limitations. No parental data; no BAM/CRAM/FASTQ for read-backed phase; no functional assay of N1002K; targeted candidate-gene analysis is not equivalent to exhaustive clinical genome interpretation. Findings are research hypotheses and not for patient care.

AI/data-handling disclosure. OpenAI ChatGPT supported public-literature discovery, synthesis, drafting, methodology/code planning, and targeted review of reduced/candidate-level VCF-derived data. Raw BAM/CRAM and the full VCF are not reproduced in this report. Restricted genomic data should not be redistributed publicly.